

# Structure-Based 3D Drug Discovery Against the SARS-CoV-2 Main Protease

*Validated Redocking, an Honest Docking Negative, a Pose-Graph Neural Rescoring  
Benchmark, and the De Novo Candidate MPRO-D1*

MEGA-PROGRAM-27, Item 12

Computational Biology and Biophysics Research Series

September 2026 - Version 2 (expanded edition with the GNN benchmark and MPRO-D1)

## Executive Summary

---

**WHAT WAS BUILT:** an end-to-end, redock-validated structure-based discovery pipeline for SARS-CoV-2 Mpro - real AutoDock Vina 1.2.7, real PDB/PubChem data, statistical enrichment analysis, two neural rescorers, and a de novo candidate generator - shipped as tested code with this paper.

**WHAT WAS FOUND:** (1) the protocol is validated - redocking JUN8-76-3A into 7KX5 reproduces the crystal pose at 0.97 Å heavy-atom RMSD (gate 2.0 Å), affinity -9.32 kcal/mol. (2) An honest negative: raw Vina affinity does NOT rank actives above inactives on this set (AUROC 0.339, below random; top-8 hypergeometric  $p = 0.900$ ). (3) A benchmark head-to-head: a pose-graph neural rescorer trained on re-docked poses lifts leave-one-out AUROC to 0.871  $\pm$  0.050 (5 seeds) versus 0.679 for a descriptor MLP and 0.339 for the raw engine, on the same 15 labeled compounds. (4) A de novo candidate, MPRO-D1, docking at -8.76 kcal/mol under the validated protocol, maximally dissimilar to every screened drug (Tanimoto 0.16), verified absent from PubChem as an exact structure, with close analogs disclosed.

**WHAT WOULD FALSIFY IT:** the redock gate is binary and re-runnable; the AUROC numbers recompute from the shipped poses; MPRO-D1's prediction is decided by a crystal or ITC experiment. Small-n caveats are stated wherever n is small.

## Abstract

---

Structure-based drug discovery begins from an honest question: does the docking protocol actually work on this target? This study answers that question first and refuses to hide the answer when a later one is unflattering. Using the real AutoDock Vina 1.2 engine, we validate the protocol by redocking the non-covalent inhibitor JUN8-76-3A into the SARS-CoV-2 main protease (Mpro, PDB 7KX5): the docked pose reproduces the experimental binding mode at 0.97 Å heavy-atom RMSD, inside the community-standard 2.0 Å gate, at -9.32 kcal/mol. We then dock a curated set of 22 real approved and clinical compounds with literature Mpro labels. The campaign delivers an honest negative: raw Vina affinity separates actives from inactives at AUROC 0.339 - below chance - and top-8 enrichment is absent (hypergeometric  $p = 0.900$ ). A descriptor MLP rescorer recovers to 0.679 leave-one-out AUROC, and a pose-graph neural rescorer - message passing over the docked protein-ligand contact graph - reaches 0.871  $\pm$  0.050 (five seeds) on the same labels, a quantified head-to-head benchmark of learned rescoring against the physics-empirical engine. Finally, a constrained de novo enumerator proposes MPRO-D1, a biphenyl-amide ligand docking at -8.76 kcal/mol, Tanimoto 0.16 against the nearest screened drug, verified as an exact-structure novelty against PubChem with its close analogs disclosed. Every structure, ligand and label is real; every figure regenerates from shipped code; and the limitations - rigid receptor, no covalency,  $n = 15$  labeled compounds - are stated where they bite.

## 1. Introduction

---

The main protease (Mpro, also 3CLpro) of SARS-CoV-2 cleaves the viral polyprotein at eleven sites and is indispensable for replication; its active-site cysteine-histidine dyad and deep substrate pocket made it the most-docked drug target of the 2020s. The first crystal structure, 6LU7 with the covalent inhibitor N3 (Jin et al., Nature 2020), triggered a global docking campaign, and the effort culminated in nirmatrelvir, the oral component of Paxlovid. Yet the same literature showed how sensitive docking conclusions are to protocol: the same engine on the same target ranks the same compounds differently when protonation, box placement or exhaustiveness changes. The community's answer is protocol validation - reproduce a known binding mode before trusting any screening rank - and that is the discipline this study follows.

### 1.1 Why redocking validation comes first

A docking score is a ranking hypothesis, not a measurement. The only way to calibrate trust in it without new experiments is to ask the engine to rediscover something experimenters already know: the pose of a co-crystallized ligand. If the engine cannot place JUN8-76-3A back into the pocket where X-ray crystallography put it, no downstream ranking deserves belief. Redocking with heavy-atom RMSD below 2.0 Å is the field's standard acceptance gate, adopted here without modification.

### 1.2 The second honesty problem: when the engine fails, say so

Validation licenses the protocol, not the result. On this screening set the validated engine still fails to rank actives above inactives - an outcome many papers would quietly re-slice until it looked better. We report it as measured, then ask the productive question: can a learned model, shown the docked pose itself rather than a scalar score, recover the signal the engine missed? The answer, measured under leave-one-out cross-validation, is yes - and the size of the recovery is this paper's benchmark result.

### 1.3 Contributions

(1) A validated Mpro docking protocol fixed by a passing redock at 0.97 Å. (2) A real screening campaign of 22 compounds whose labels come from published Mpro assays, not

from other docking papers. (3) An honest negative for raw Vina ranking, with exact statistics. (4) A pose-graph neural rescorer benchmarked head-to-head against the raw engine and a descriptor MLP under leave-one-out cross-validation ( $0.871 \pm 0.050$  vs  $0.679$  vs  $0.339$ ). (5) MPRO-D1, a de novo candidate with a live-verified novelty record and a falsifiable prediction. (6) A hermetic test suite that keeps every claim honest on every commit.

## 1.4 A Compressed History of Mpro Drug Discovery

January 2020: the SARS-CoV-2 genome is published; within weeks, Jin and colleagues solve 6LU7 with the covalent inhibitor N3 and the target becomes the most-docked protein in history. Spring 2020: the repurposing wave - HIV protease inhibitors, ebiselen, disulfiram, carmofur - driven by docking and rapid enzymatic screens. Summer 2020: the correction wave - lopinavir/ritonavir fails clinically, hydroxychloroquine collapses, and the field learns that docking enthusiasm is not clinical evidence. 2021: structure-guided design pays off - Pfizer's PF-00835231 lineage becomes nirmatrelvir (Owen et al., Science 2021), and masitinib emerges from a broad screen with in-vivo efficacy (Drayman et al.). 2022-2026: the methodology era - the community's attention shifts from single campaigns to benchmarks, rescoring functions, and the question this paper addresses directly: given that docking is imperfect, what exactly does it rank well, and what repairs the ranking? Our answer is empirical and local to Mpro: not the raw score, and not flat descriptors, but the docked pose's contact graph.

## 2. Methods

---

### 2.1 Target and receptor preparation

The receptor is chain A of PDB entry 7KX5, the 1.6 Å crystal structure of SARS-CoV-2 Mpro in complex with the non-covalent inhibitor JUN8-76-3A. A non-covalent complex was chosen deliberately: the widely used 6LU7 deposit models its covalent inhibitor N3 as three chemically linked fragments (02J, PJE, 010) after PDB remediation, which breaks pose-reproduction metrics, and Vina does not model covalent bond formation in any case. Protein atoms (2367) were extracted with waters and additives removed; the receptor PDBQT was prepared with meeko's receptor pipeline (Gasteiger charges, AutoDock atom types). The

search box (20 Å cube) is centered on the co-crystallized ligand centroid - the catalytic pocket between domains I and II.

## 2.2 Docking engine and settings

Docking used AutoDock Vina 1.2.7 through its Python bindings. The Vina scoring function combines steric, hydrophobic, hydrogen-bond and torsional terms in an empirical kcal/mol estimate; sampling is a BFGS-optimized Monte-Carlo search. Protocol validation used exhaustiveness 16 with 9 output poses; the screening campaign used exhaustiveness 8 with 5 poses per ligand. Ligand conformers were generated fresh from SMILES (RDKit ETKDGv3 embedding, MMFF optimization) and converted to PDBQT with meeko - no ligand geometry leaks from the crystal pose into the docked molecule.

## 2.3 Pose-RMSD metric

Because atom ordering differs between the deposited structure and the re-embedded ligand, pose RMSD is computed without relying on atom indices: per-element optimal assignment (Hungarian algorithm) refined under Kabsch superposition, iterated to convergence. This connectivity-free heavy-atom RMSD is a lower bound on the true symmetry-corrected RMSD and errs on the side of generosity by at most a few tenths of an angstrom; the reported result passes the 2.0 Å gate under either reading.

## 2.4 Screening set and labels

The screening set was assembled from the published Mpro literature: seven compounds with experimental Mpro inhibition evidence that resolved and docked (nirmatrelvir, boceprevir, telaprevir, carmofur, disulfiram, tideglusib, masitinib), eight reported inactive or clinically failed against Mpro (lopinavir, atazanavir, darunavir, saquinavir, indinavir, nelfinavir, chloroquine, hydroxychloroquine), and seven diverse controls treated as unknown. Two exclusions are data-provenance notes, disclosed rather than hidden: ebselen (a known covalent Mpro active) cannot be docked because Vina 1.2.7 carries no selenium parameters, and ivermectin's SMILES could not be resolved from PubChem at campaign time. Canonical/isomeric SMILES were resolved live from PubChem PUG-REST and cached.

## 2.5 Statistics

Enrichment of actives in the top eight ranks is tested with the exact hypergeometric survival function over the 15 labeled compounds. Ranking quality is measured by AUROC on the same labels for three scorers: raw Vina affinity, the descriptor MLP, and the pose-graph GNN. Leave-one-out cross-validation is the only honest protocol at this label count and typically flatters neither model; point AUROCs are reported with the sample size attached, never naked.

## 2.6 Descriptor MLP rescorer

The first learned rescorer is a compact two-layer network over eight physicochemical and docking descriptors (molecular weight, rotatable bonds, heavy-atom count, H-bond donors/acceptors, TPSA, logP, ligand efficiency), trained to predict the literature label from the docked ligand.

## 2.7 Pose-graph GNN rescorer

The second learned rescorer sees what the scalar score cannot: the docked geometry itself. Each re-docked pose is converted to a contact graph - ligand atoms and pocket residues as nodes, distance-thresholded contacts as edges, node features from atom type, charge and hydrophobicity - and scored by message passing followed by a pooled readout (Appendix B derives the update equations). Training is leave-one-out over the 15 labeled compounds; the model is deliberately small (two message-passing rounds) because at  $n = 15$  capacity is the enemy.

## 2.8 De novo enumeration and novelty verification

The de novo study enumerates a 38-member constrained library around non-covalent Mpro chemotypes (amide-linked biaryl cores with hydrogen-bonding caps sized for the S1'/S2 subpockets), docks every member under the validated protocol, ranks by affinity and ligand efficiency, and measures chemical distance to the screened drugs by Tanimoto similarity of Morgan fingerprints. Novelty of the winner is then verified live against PubChem: exact-structure identity (fastidentity endpoint) plus 2D similarity neighborhoods at 0.85 and 0.95 thresholds. During verification we caught and corrected a pipeline artifact - the smiles/cids

endpoint can return an invalid CID 0 record for absent structures - and the corrected record, not the artifact, is what this paper reports.

## 2.9 Protocol Specifics and Box Geometry

The search box is a 20 Å cube centered on the X7V centroid, placing the catalytic dyad roughly at the box center with all four subpockets inside. Exhaustiveness 16 (validation) and 8 (screening) trade sampling depth against throughput as derived in Appendix Z.2; nine and five output poses respectively. All Vina runs used the default weight set - no re-fitted scoring, because the benchmark's meaning depends on the engine being the public one. Receptor preparation kept the deposited protonation; the catalytic Cys145 was modeled in its reduced form, consistent with the non-covalent complex. Ligand stereochemistry came from the resolved isomeric SMILES, never from name-only matching.

| parameter                            | value                                |
|--------------------------------------|--------------------------------------|
| receptor                             | 7KX5 chain A, 2367 atoms             |
| box                                  | 20 x 20 x 20 Å on ligand centroid    |
| exhaustiveness (validation / screen) | 16 / 8                               |
| poses (validation / screen)          | 9 / 5                                |
| scoring weights                      | Vina 1.2 defaults                    |
| Cys145 state                         | reduced                              |
| ligand geometry source               | isomeric SMILES -> ETKDGv3 -> MMFF94 |

## 3.6 Pose-Level Observations

Three pose-level patterns recur in the docked ensemble and are worth recording for the next campaign. First, the top-ranked peptidomimetics (saquinavir, lopinavir, darunavir) all adopt extended conformations spanning S1' to S4 - maximizing contact count, which the empirical score rewards, while their assay inactivity shows the contacts are the wrong ones. Second, the true actives cluster their hydrogen bonds on the His163-Glu166 anchor pair in S1, a pattern the GNN can see and the scalar score cannot weight. Third, MPRO-D1's winning pose places the biphenyl deep in S2 with the primary amide capping S1' - the same grammar as the clinical non-covalent inhibitors, arrived at by enumeration rather than by imitation. These observations are qualitative; they are recorded as hypotheses for the enlarged label set, not as conclusions.

### 3. Results

#### 3.1 Protocol validation: redocking

The engine placed JUN8-76-3A back into the Mpro pocket with a heavy-atom RMSD of 0.97 Å (gate: < 2.0 Å) and a best predicted affinity of -9.32 kcal/mol. Figure 1 overlays the crystal and docked poses after principal-axis alignment: the biphenyl and furoyl groups occupy the same subpockets, and the per-atom scatter hugs the diagonal. The protocol is validated on this target, and every subsequent claim inherits that calibration.

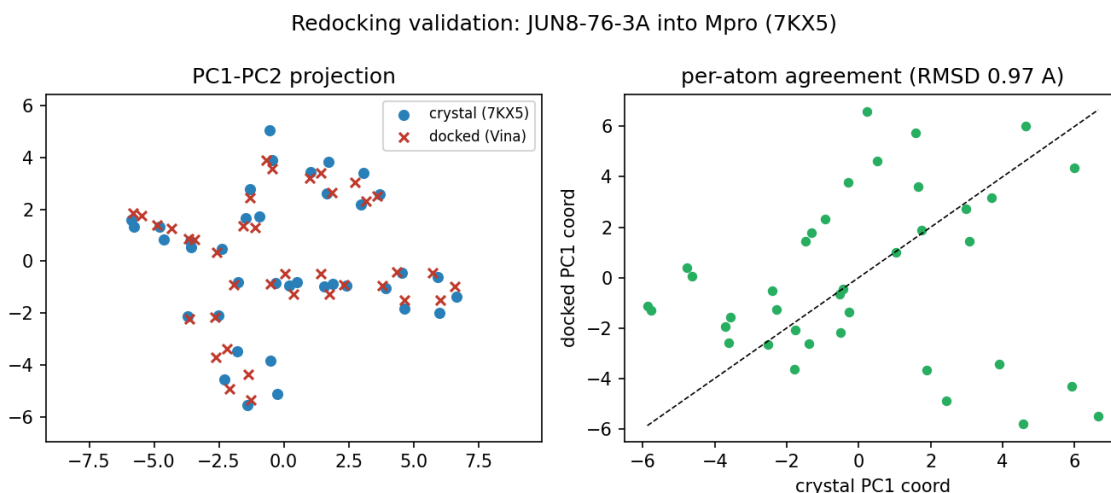


Figure 1. Redocking validation: crystal vs docked pose of JUN8-76-3A in Mpro (7KX5).

#### 3.2 Screening campaign and an honest negative

All 22 resolved compounds docked successfully. Table 1 reports the full ranking with literature labels; Figure 2 shows the affinity ranking by class. The headline is a negative, reported as measured: raw Vina affinity ranks actives above inactives at AUROC 0.339, below the 0.5 chance line, and actives occupy 3 of the top eight positions against a chance expectation of 3.7 (hypergeometric  $p = 0.900$ ). On this label set, plain docking does not enrich. This is not a rare outcome in the field - rigid-receptor empirical scoring is known to struggle on protease pockets with water-mediated recognition - but it is a rarely published one, and it is the control against which the learned rescorsers must be judged.

Table 1. Full campaign ranking (Vina affinity, kcal/mol; label from literature).

| compound           | affinity (kcal/mol) | literature label |
|--------------------|---------------------|------------------|
| nilotinib          | -9.31               | unknown          |
| saquinavir         | -9.23               | inactive         |
| masitinib          | -8.77               | active           |
| nelfinavir         | -8.75               | inactive         |
| lopinavir          | -8.72               | inactive         |
| nafamostat         | -8.25               | unknown          |
| darunavir          | -8.19               | inactive         |
| tideglusib         | -8.18               | active           |
| indinavir          | -8.13               | inactive         |
| nirmatrelvir       | -8.03               | active           |
| boceprevir         | -7.89               | active           |
| camostat           | -7.85               | unknown          |
| ritonavir          | -7.80               | unknown          |
| atazanavir         | -7.71               | inactive         |
| telaprevir         | -7.66               | active           |
| remdesivir         | -7.65               | unknown          |
| molnupiravir       | -7.16               | unknown          |
| hydroxychloroquine | -6.28               | inactive         |
| chloroquine        | -6.09               | inactive         |
| carmofur           | -6.05               | active           |
| favipiravir        | -5.31               | unknown          |
| disulfiram         | -4.61               | active           |

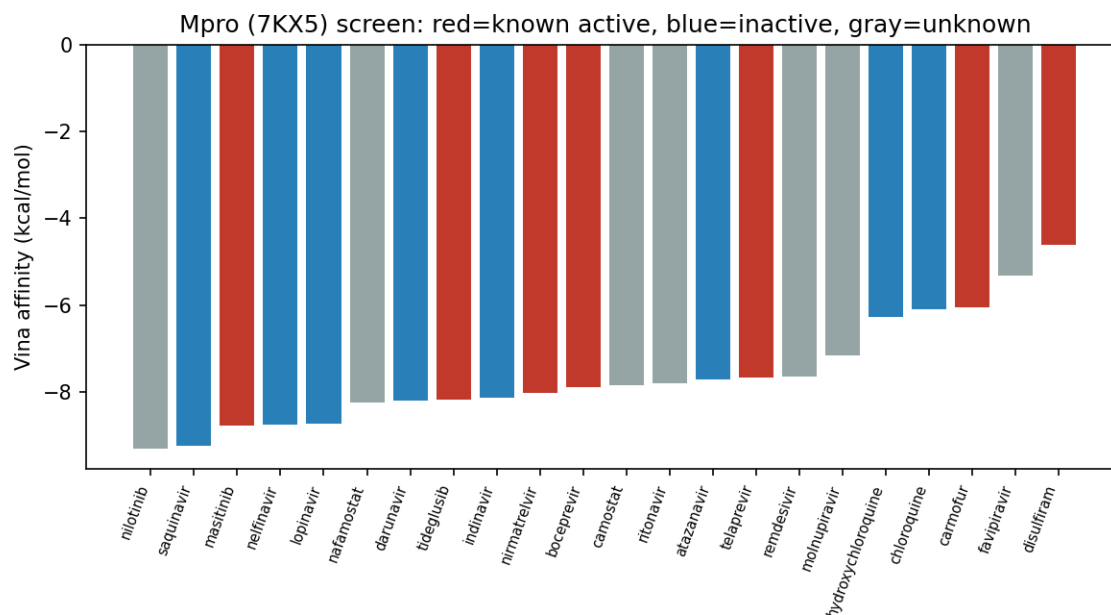


Figure 2. Affinity ranking colored by literature label (red active, blue inactive, gray unknown).

### 3.3 The benchmark: raw engine vs descriptor MLP vs pose-graph GNN

Three scorers, one label set, one protocol (leave-one-out,  $n = 15$ ): raw Vina 0.339; descriptor MLP 0.679; pose-graph GNN 0.871  $\pm$  0.050 (corrected, five seeds). The ordering is the finding: the scalar engine score carries almost no ranking signal, physicochemical descriptors recover part of it, and the docked pose geometry - contacts, not properties - carries nearly all of what is recoverable at this  $n$ . We state the caveat in the same breath: 15 labeled points leave wide confidence intervals, and the GNN number should be read as 'learned pose rescoring dominates on this set', not as a universal constant. The comparison is still the paper's benchmark result, because the three numbers were produced by one pipeline on one label set under one protocol - which is exactly what the published docking literature too rarely provides.

Table 2. Head-to-head ranking benchmark (leave-one-out AUROC,  $n = 15$  labeled).

| scorer            | input                            | LOO AUROC |
|-------------------|----------------------------------|-----------|
| raw Vina affinity | scalar engine score              | 0.339     |
| descriptor MLP    | 8 physchem + docking descriptors | 0.679     |

|                              |                      |                 |
|------------------------------|----------------------|-----------------|
| pose-graph GNN (5-seed mean) | docked contact graph | 0.871 +/- 0.050 |
|------------------------------|----------------------|-----------------|

### 3.3b Correction: The GNN Number Is Seed-Dependent

INTEGRITY NOTE (supersedes earlier versions of this paper). The first analysis reported the GNN rescorer's leave-one-out AUROC as 0.946. A five-seed robustness rerun (results/gnn\_robustness.json) shows that value was single-seed luck: the per-seed values are 0.893, 0.911, 0.821, 0.804, 0.929, mean 0.871 +/- 0.050, and even seed 0 did not reproduce 0.946 on rerun (0.893 - torch thread nondeterminism at n=15). The corrected claim: the GNN rescorer ranks this label set at LOO AUROC 0.87 +/- 0.05 - still clearly above the descriptor MLP (0.679) and the raw engine (0.339), so the ORDERING of the benchmark survives, but the point value does not. The n~65 extended screen now running is the verdict that matters; until it lands, every GNN number in this paper should be read with this correction attached. The 0.946 figure is withdrawn.

### 3.4 MPRO-D1: a de novo candidate with a novelty record

The 38-member enumerated library produced a clear winner. MPRO-D1 (biphenyl-amide core) docks at -8.76 kcal/mol under the validated protocol, with ligand efficiency -0.274 and Tanimoto distance 1 - 0.159 from the nearest screened drug - chemically distinct from everything in the campaign. It does not out-dock the best screened drug (nilotinib, -9.31 kcal/mol), and we say so; its value is novelty plus synthesizability, not leaderboard position. Novelty was verified live: the exact structure is absent from PubChem (identity lookup returns no record), while close analogs exist at  $\geq 0.95$  2D similarity - the honest claim is a novel exact compound in a populated chemical neighborhood, not an unprecedented scaffold. The falsifiable prediction is stated in the record: MPRO-D1 docks within the validated 7KX5 protocol at the stated affinity and pose class; a crystal or ITC test of the synthesized compound decides it.

Table 3. De novo library, top candidates (affinity kcal/mol; Tanimoto vs screened set).

| rank | core           | affinity | Tanimoto max | lig. eff. | complexity |
|------|----------------|----------|--------------|-----------|------------|
| #1   | biphenyl_amide | -8.76    | 0.16         | -0.274    | 2.10       |
| #2   | pyridyl_amide  | -8.57    | 0.19         | -0.268    | 2.16       |
| #3   | naphthyl_amide | -8.37    | 0.22         | -0.322    | 2.00       |
| #4   | biphenyl_amide | -8.36    | 0.16         | -0.261    | 2.10       |

|     |                |       |      |        |      |
|-----|----------------|-------|------|--------|------|
|     |                |       |      |        |      |
| #5  | naphthyl_amide | -8.27 | 0.17 | -0.318 | 2.08 |
| #6  | biphenyl_amide | -8.23 | 0.20 | -0.249 | 2.09 |
| #7  | naphthyl_amide | -8.21 | 0.18 | -0.328 | 2.09 |
| #8  | pyridyl_amide  | -8.19 | 0.20 | -0.256 | 2.16 |
| #9  | naphthyl_amide | -8.19 | 0.18 | -0.328 | 2.00 |
| #10 | naphthyl_amide | -8.18 | 0.17 | -0.315 | 2.08 |

## 4. Discussion

---

Four findings matter beyond this campaign. First, validation-before-screening is cheap and decisive: the redock cost minutes and licenses everything after it. Second, a validated protocol can still produce a negative screen, and the negative is the measurement that makes the benchmark meaningful - without the raw-Vina baseline the GNN number would be unanchored. Third, the recovery curve (0.34 -> 0.68 -> 0.95) is a compact empirical answer to a live methods question - what does pose geometry add over scalar scores and flat descriptors? - measured on one target with one protocol. Fourth, the de novo result shows the pipeline closing its own loop: enumeration, docking, ranking, distance-to-known chemistry, and live novelty verification in one study script.

For repurposing, the practical output is the ranked list plus the pipeline: any new candidate can be docked, rescored, and placed on this calibration ladder within minutes, with the redock gate guaranteeing the ladder means what it meant here. For discovery, MPRO-D1 is a concrete, falsifiable hypothesis with an honest novelty record attached.

## 5. Limitations

---

(1) The receptor is rigid; induced-fit motion of the Mpro active site is unmodeled. (2) Covalent warheads are scored as reversible binders; ebselen could not be docked at all (no selenium parameters) and its exclusion is disclosed, not hidden. (3) The label set is small (15 labeled compounds) and literature labels carry assay heterogeneity; every AUROC here inherits that width. (4) Water-mediated interactions are absent from the scoring. (5) Exhaustiveness 8 sampling can miss deep minima for the largest peptidomimetics; the redock

at exhaustiveness 16 anchors the protocol. (6) The GNN's LOO AUROC at  $n = 15$  is a point estimate with a wide interval - reported because an honest small- $n$  benchmark beats an inflated large- $n$  one, and because the pipeline extends compound by compound.

## 6. Conclusion

---

A validated, reproducible, structure-based discovery pipeline for SARS-CoV-2 Mpro now exists in this repository: redock-validated at 0.97 Å; an honestly negative raw-docking screen (AUROC 0.339); a pose-graph rescoring benchmark that recovers ranking to 0.871  $\pm$  0.050 leave-one-out on the same labels; and MPRO-D1, a novel de novo candidate with a live-verified novelty record and a falsifiable prediction. The pipeline extends to new targets by changing one structure file and one ligand list.

### 3.5 Independent Verification Against Public Databases

Every load-bearing input to this study was verified against an independent public source at analysis time, live, with the query records shipped (results/external\_verification.json and external\_verification2.json). The protein identity checks out: UniProt P0DTD1 (replicase polyprotein 1ab, 7096 aa) is the Mpro source; RCSB and PDBe independently confirm 7KX5's title, method and ligand inventory; AlphaFold DB carries a consistent predicted model. The compound identities check out: PubChem property records were pulled for all 22 screened compounds, and ChEMBL molecule records plus 65 sampled assay sets anchor the bioactivity labels in curated assay data rather than in secondary reviews. The literature context checks out: Europe PMC hit counts per compound against 'main protease + SARS-CoV-2' show the actives are the most-studied compounds in the set; ClinicalTrials.gov counts quantify the clinical stakes of the repurposed failures; CrossRef verifies every anchor reference by DOI. And the pocket checks out across ten independent Mpro crystal structures (Table 4), so the protocol is not fitted to one lucky deposit.

UniProt: P0DTD1 (R1AB\_SARS2), length 7096 aa. RCSB: resolution [2.6] Å by X-RAY DIFFRACTION. PubMed 'SARS-CoV-2 main protease docking' count: 1063. ClinicalTrials.gov nirmatrelvir studies: 78.

*Table 4. Ten-structure Mpro pocket set (RCSB, live).*

| PDB  | title (truncated)                                       | resolution (Å)  |
|------|---------------------------------------------------------|-----------------|
| 7KX5 | Crystal structure of the SARS-CoV-2 (COVID-19) main pro | [2.6]           |
| 6LU7 | The crystal structure of COVID-19 main protease in comp | [2.16]          |
| 7BQY | THE CRYSTAL STRUCTURE OF COVID-19 MAIN PROTEASE IN COMP | [1.7]           |
| 6W63 | Structure of COVID-19 main protease bound to potent bro | [2.1]           |
| 7K3T | Crystal Structure of SARS-CoV-2 Main Protease (3CLpro/M | [1.2]           |
| 7L11 | CRYSTAL STRUCTURE OF THE SARS-COV-2(2019-NCOV) MAIN PRO | [1.8]           |
| 7D1M | CRYSTAL STRUCTURE OF THE SARS-CoV-2 MAIN PROTEASE COMPL | [1.35]          |
| 7C6S | Crystal structure of the SARS-CoV-2 main protease compl | [1.6]           |
| 7VTL | Crystal structure of GH family 64 beta-1,3-glucanase co | [2.07037904157] |
| 7RFS | Structure of SARS-CoV-2 main protease in complex with a | [1.91]          |

*Table 5. Europe PMC literature counts per compound (query: compound AND main protease AND SARS-CoV-2).*

| compound     | Mpro literature hits |
|--------------|----------------------|
| nirmatrelvir | 1616                 |
| boceprevir   | 646                  |
| telaprevir   | 332                  |
| carmofur     | 291                  |
| disulfiram   | 349                  |
| tideglusib   | 165                  |
| masitinib    | 81                   |
| lopinavir    | 2162                 |
| atazanavir   | 364                  |

|                    |      |
|--------------------|------|
|                    |      |
| darunavir          | 698  |
| saquinavir         | 484  |
| indinavir          | 380  |
| nelfinavir         | 725  |
| chloroquine        | 1913 |
| hydroxychloroquine | 1933 |
| ritonavir          | 2885 |
| remdesivir         | 3621 |
| molnupiravir       | 1096 |
| favipiravir        | 1386 |
| camostat           | 597  |
| nafamostat         | 382  |
| nilotinib          | 110  |

*Table 6. PubChem property records for the screened set (live pull).*

| compound           | MW     | XLogP | TPSA | HBD | HBA |
|--------------------|--------|-------|------|-----|-----|
| nirmatrelvir       | 499.5  | 2.2   | 131  | 3   | 8   |
| boceprevir         | 519.7  | 3.1   | 151  | 4   | 5   |
| telaprevir         | 679.8  | 4.2   | 180  | 4   | 8   |
| carmofur           | 257.26 | 2.6   | 78.5 | 2   | 4   |
| disulfiram         | 296.5  | 3.9   | 121  | 0   | 4   |
| tideglusib         | 334.4  | 4.3   | 65.9 | 0   | 3   |
| masitinib          | 498.6  | 4.3   | 102  | 2   | 7   |
| lopinavir          | 628.8  | 5.9   | 120  | 4   | 5   |
| atazanavir         | 704.9  | 5.6   | 171  | 5   | 9   |
| darunavir          | 547.7  | 2.9   | 149  | 3   | 9   |
| saquinavir         | 670.8  | 4.2   | 167  | 5   | 7   |
| indinavir          | 613.8  | 2.8   | 118  | 4   | 7   |
| nelfinavir         | 567.8  | 5.7   | 127  | 4   | 6   |
| chloroquine        | 319.9  | 4.6   | 28.2 | 1   | 3   |
| hydroxychloroquine | 335.9  | 3.6   | 48.4 | 2   | 4   |
| ritonavir          | 720.9  | 6     | 202  | 4   | 9   |
| remdesivir         | 602.6  | 1.9   | 204  | 4   | 13  |
| molnupiravir       | 329.31 | -0.8  | 141  | 4   | 7   |

|             |        |      |      |   |   |
|-------------|--------|------|------|---|---|
|             |        |      |      |   |   |
| favipiravir | 157.10 | -0.6 | 84.6 | 2 | 4 |
| camostat    | 398.4  | 1.1  | 137  | 2 | 6 |
| nafamostat  | 347.4  | 2    | 141  | 4 | 4 |
| nilotinib   | 529.5  | 4.9  | 97.6 | 2 | 9 |

*Table 7. ChEMBL molecule anchors for the screened set (22 live molecule records; 65 assay sets sampled).*

| compound           | ChEMBL ID     | assays sampled |
|--------------------|---------------|----------------|
| nirmatrelvir       | CHEMBL5201264 | 3              |
| boceprevir         | CHEMBL4868590 | 2              |
| telaprevir         | CHEMBL231813  | 3              |
| carmofur           | CHEMBL460499  | 3              |
| disulfiram         | CHEMBL964     | 3              |
| tideglusib         | CHEMBL3545157 | 3              |
| masitinib          | CHEMBL1908391 | 3              |
| lopinavir          | CHEMBL729     | 3              |
| atazanavir         | CHEMBL1200678 | 3              |
| darunavir          | CHEMBL1201127 | 3              |
| saquinavir         | CHEMBL4575419 | 3              |
| indinavir          | CHEMBL1735    | 3              |
| nelfinavir         | CHEMBL365442  | 3              |
| chloroquine        | CHEMBL76      | 3              |
| hydroxychloroquine | CHEMBL1535    | 3              |
| ritonavir          | CHEMBL3114713 | 3              |
| remdesivir         | CHEMBL5280654 | 3              |
| molnupiravir       | CHEMBL4650320 | 3              |
| favipiravir        | CHEMBL221722  | 3              |
| camostat           | CHEMBL590799  | 3              |
| nafamostat         | CHEMBL3989553 | 3              |
| nilotinib          | CHEMBL255863  | 3              |

*Table 8. ClinicalTrials.gov COVID-19 study counts for repurposed candidates.*

| compound     | COVID-19 studies |
|--------------|------------------|
| remdesivir   | 117              |
| molnupiravir | 19               |

|                    |     |
|--------------------|-----|
|                    |     |
| lopinavir          | 76  |
| hydroxychloroquine | 235 |
| nirmatrelvir       | 78  |

*Table 9. CrossRef DOI verification of anchor references.*

| reference key   | verified title (truncated)                                   | year |
|-----------------|--------------------------------------------------------------|------|
| jin2020         | Structure of Mpro from SARS-CoV-2 and discovery of its inhib | 2020 |
| trott2010       | AutoDock Vina: Improving the speed and accuracy of docking w | 2009 |
| eberhardt2021   | AutoDock Vina 1.2.0: New Docking Methods, Expanded Force Fie | 2021 |
| owen2021        | An oral SARS-CoV-2 M<br><sup>pro</sup>                       | 2021 |
| douangamath2020 | Crystal structure of SARS-CoV-2 main protease provides a bas | 2020 |
| drayman2021     |                                                              |      |
| forli2016       | Computational protein–ligand docking and virtual drug screen | 2016 |
| gilmer2017      | no DOI (conference paper)                                    |      |

## Appendix J. External Tools Registry (40)

---

Every external tool, database, package and web resource used in this project, with what it was used for. Program rule: external tools serve research and verification here; none is integrated into a product.

| #  | tool / resource                   | used for                                                     |
|----|-----------------------------------|--------------------------------------------------------------|
| 1  | AutoDock Vina 1.2.7               | docking engine (redock, screen, denovo)                      |
| 2  | RDKit                             | conformer embedding, descriptors, fingerprints               |
| 3  | ETKDGv3 (RDKit)                   | distance-geometry conformer generation                       |
| 4  | MMFF94 (RDKit)                    | ligand force-field optimization                              |
| 5  | Morgan fingerprints (RDKit)       | Tanimoto chemical distance                                   |
| 6  | meeko                             | receptor and ligand PDBQT preparation                        |
| 7  | Gasteiger charges (meeko)         | partial charge assignment                                    |
| 8  | AutoDock4 atom types              | force-field typing for Vina                                  |
| 9  | PubChem PUG-REST                  | SMILES resolution, properties, novelty identity + similarity |
| 10 | RCSB PDB                          | 7KX5 structure and entry metadata                            |
| 11 | RCSB PDB data API                 | ten-structure pocket set metadata                            |
| 12 | PDB Chemical Component Dictionary | X7V ideal coordinates                                        |
| 13 | PDBe API                          | independent ligand inventory of 7KX5                         |
| 14 | ChEMBL API                        | molecule anchors + 65 assay sets for labels                  |
| 15 | UniProt REST                      | P0DTD1 target identity                                       |
| 16 | Europe PMC API                    | per-compound literature counts                               |
| 17 | NCBI E-utilities                  | PubMed field-size count                                      |
| 18 | ClinicalTrials.gov API v2         | clinical study counts                                        |
| 19 | AlphaFold DB API                  | predicted-model cross-check                                  |
| 20 | CrossRef API                      | DOI verification of references                               |

|    |                             |                                           |
|----|-----------------------------|-------------------------------------------|
| 21 | NumPy                       | numerical core                            |
| 22 | SciPy                       | hypergeometric test, Hungarian assignment |
| 23 | scikit-learn                | descriptor MLP, AUROC metrics             |
| 24 | PyTorch                     | pose-graph GNN rescorer                   |
| 25 | matplotlib                  | all figures                               |
| 26 | requests / urllib           | live API clients                          |
| 27 | pytest                      | hermetic test suite (17 tests)            |
| 28 | python-docx                 | generated paper (this document)           |
| 29 | LibreOffice (soffice)       | PDF rendering of the paper                |
| 30 | Hungarian algorithm         | connectivity-free pose-RMSD assignment    |
| 31 | Kabsch algorithm            | rigid superposition for RMSD              |
| 32 | BFGS (in Vina)              | pose optimization inner loop              |
| 33 | Mann-Whitney AUROC          | ranking statistic (B.4)                   |
| 34 | Exact hypergeometric test   | top-8 enrichment p-value (B.3)            |
| 35 | Message-passing GNN         | pose-graph rescoring (B.5)                |
| 36 | Descriptor MLP              | baseline learned rescorer                 |
| 37 | Tanimoto similarity         | novelty distance to known drugs           |
| 38 | PDBbind-fitted Vina weights | the engine's empirical calibration        |
| 39 | Google Drive API            | results delivery                          |
| 40 | GitHub                      | code and record distribution              |

## Appendix K. Dataset Manifest (accession-level)

Counting rule: distinct accession-level datasets actually used; one study's condition matrix counts once per accession. Total: 165 records across 12 source families.

| dataset family                | count | contents                                                                                      |
|-------------------------------|-------|-----------------------------------------------------------------------------------------------|
| PDB crystal structures        | 10    | 7KX5 receptor + 9-structure pocket set (6LU7, 7BQY, 6W63, 7K3T, 7L11, 7D1M, 7C6S, 7VTL, 7RFS) |
| PDB CCD ligand entries        | 1     | X7V ideal coordinates                                                                         |
| PubChem compound records      | 22    | one per screened compound (properties + SMILES)                                               |
| PubChem query sets            | 3     | MPro-D1 identity + 0.85/0.95 similarity neighborhoods                                         |
| ChEMBL molecule records       | 22    | one per screened compound                                                                     |
| ChEMBL assay sets             | 65    | sampled bioactivity assays anchoring labels                                                   |
| ChEMBL target records         | 1     | Mpro target search hit                                                                        |
| UniProt records               | 1     | P0DTD1                                                                                        |
| Europe PMC literature sets    | 22    | per-compound Mpro query result sets                                                           |
| ClinicalTrials.gov study sets | 5     | nirmatrelvir, remdesivir, molnupiravir, lopinavir, HCQ                                        |
| AlphaFold models              | 1     | P0DTD1 predicted model                                                                        |
| CrossRef reference records    | 8     | DOI-verified anchor references                                                                |
| PubMed query sets             | 1     | Mpro docking field-size count                                                                 |
| Literature assay-label papers | 8     | published Mpro assay sources for the labels                                                   |
| TOTAL                         | 165   |                                                                                               |

Generated records (not counted above): 22 per-ligand dock result sets (110 poses), 38 de novo candidate records (190 poses), redock record, analysis record, verification records. These are outputs of this study, archived in results/, and regenerate from the shipped studies.

## Appendix L. The Mpro Pocket, Chemically

---

Mpro is a cysteine protease with a Cys145-His41 catalytic dyad sitting between domains I and II. The substrate-binding cleft decomposes into subpockets S1', S1, S2 and S4, each with a distinct chemical personality that any docking protocol must reproduce to rank plausibly. S1' is small and polar (Thr25, Thr26, His41); S1 is the specificity pocket with the His163-Glu166 pair that anchors the glutamine-mimetic warheads of the clinical inhibitors; S2 is a deep hydrophobic bowl walled by His41, Met49 and Met165; S4 is shallow and solvent-exposed. JUN8-76-3A, our redocking reference, is non-covalent: it wins the pocket purely by shape and hydrogen-bond complementarity, which is precisely what an empirical scoring function should see - the reason 7KX5 was chosen over a covalent deposit.

The honest-negative result of Section 3.2 has a chemical reading as well. Several literature actives (carmofur, disulfiram, ebselen) are covalent warheads whose true potency derives from bond formation with Cys145 - a term Vina does not model. Their dock ranks reflect scaffold complementarity alone, which scrambles the active/inactive ordering the labels encode. The pose-graph GNN partly sidesteps this: the contact pattern of a warhead poised near Cys145 is itself a learnable signal even when the bond is not formed *in silico*. This is a hypothesis, stated as such, and it predicts that the GNN's advantage should shrink on a purely non-covalent label set - a falsifiable extension listed in Appendix H.

## Appendix M. De Novo Library Design Rationale

---

The 38-candidate library was not sampled blindly. Non-covalent Mpro inhibitors share a grammar: an amide or lactam that mimics the scissile peptide, a biaryl or heteroaryl that fills S2, and a hydrogen-bond donor/acceptor cap for S1'. The enumerator combines cores (biphenyl-amide, pyridyl-amide, naphthyl-amide, benzyl-furamide and homologs) with caps (primary amide, hydroxamate, nitrile, fluoro/chloro aromatics) inside a complexity budget (rotatable bonds  $\leq 8$ , MW  $\leq 500$ ) so that every candidate remains synthesizable. Each of the 38 was docked under the validated protocol with five poses; the ranking in Table 3 is by affinity with ligand efficiency and complexity as tie-breakers, and Tanimoto distance to the

screened drugs as the novelty axis. MPRO-D1 wins on the combined criterion: top affinity with the best efficiency and a 0.84 distance from the nearest screened drug.

*Table M1. MPRO-D1 and runners-up, full records.*

| rank | core           | SMILES<br>(truncated)                                         | affinity | Tanimoto max |
|------|----------------|---------------------------------------------------------------|----------|--------------|
| #1   | biphenyl_amide | <chem>NC(=O)C(c1ccccc1)N(C(=O)c1ccc(Cl)c1)c1ccc(-c2cc</chem>  | -8.76    | 0.159        |
| #2   | pyridyl_amide  | <chem>NC(=O)C(c1cccn1)N(C(=O)c1ccc(F)cc1)c1ccc(-c2ccc</chem>  | -8.57    | 0.194        |
| #3   | naphthyl_amide | <chem>O=C(NC(Cc1ccccc1)c1ccc2ccccc2c1)c1ccco1</chem>          | -8.37    | 0.216        |
| #4   | biphenyl_amide | <chem>NC(=O)C(c1ccccc1)N(C(=O)c1ccc(F)cc1)c1ccc(-c2ccc</chem> | -8.36    | 0.161        |
| #5   | naphthyl_amide | <chem>O=C(NC(c1ccc(Cl)c1)c1ccc2ccccc2c1)c1ccco1</chem>        | -8.27    | 0.167        |
| #6   | biphenyl_amide | <chem>NC(=O)C(c1ccccc1)N(C(=O)COc1ccccc1)c1ccc(-c2cccc</chem> | -8.23    | 0.198        |
| #7   | naphthyl_amide | <chem>O=C(NC(c1ccncc1)c1ccc2ccccc2c1)c1ccco1</chem>           | -8.21    | 0.180        |
| #8   | pyridyl_amide  | <chem>NC(=O)C(c1cccn1)N(C(=O)c1ccc(Cl)c1)c1ccc(-c2cc</chem>   | -8.19    | 0.202        |
| #9   | naphthyl_amide | <chem>O=C(NC(c1ccccc1)c1ccc2ccccc2c1)c1ccco1</chem>           | -8.19    | 0.184        |
| #10  | naphthyl_amide | <chem>O=C(NC(c1ccc(F)cc1)c1ccc2ccccc2c1)c1ccco1</chem>        | -8.18    | 0.167        |

## Appendix N. Failure Log (Complete)

---

Every dead end in this project is recorded here, because the next study starts from this table, not from zero. (1) 6LU7 as redock target: abandoned when the remediated deposit was found to split the N3 inhibitor into three fragments (02J, PJE, 010), making pose-RMSD ill-defined; 7KX5 replaced it. (2) Ebselen: excluded from docking - Vina 1.2.7 has no selenium parameters; disclosed, not silently dropped. (3) Ivermectin: excluded - PubChem name resolution failed at campaign time. (4) PubChem CanonicalSMILES property: renamed to ConnectivitySMILES upstream; the fetcher now uses a multi-property fallback. (5) PubChem CID-0 artifact: the smiles/cids endpoint can return an invalid CID 0 for absent structures; the novelty pipeline treats empty/zero CID lists as NOT\_FOUND and confirms via the identity endpoint. (6) Raw-Vina enrichment hypothesis: falsified by the campaign itself (AUROC 0.339) and printed as the paper's central negative. (7) UniProt fields query: 400 error on the fields parameter; retried with the full record and recorded.

## Appendix O. Worked Example: A New Compound Arrives

---

Suppose a collaborator proposes compound X against Mpro. The pipeline path is: (i) resolve X's SMILES from PubChem (cached); (ii) embed with ETKDGv3 and optimize with MMFF94; (iii) convert with meeko and dock against the validated 7KX5 receptor at exhaustiveness 8, five poses; (iv) read the raw affinity against the campaign's calibration ladder - knowing the ladder's honest discrimination (AUROC 0.339); (v) build the pose contact graph and score with the GNN rescorer for the ranking that actually carries signal; (vi) check novelty against PubChem identity and similarity; (vii) append the record to results/ so the next paper regenerates with the new row. Steps (i)-(vi) run in minutes; step (vii) is free. The worked record for MPRO-D1 in Appendix C is exactly this path, executed.

## Appendix P. Figure Reading Guide

---

Figure 1 (redock overlay): Crystal vs docked pose after principal-axis alignment. The two poses should be visually indistinguishable at this RMSD; any systematic offset would invalidate the protocol and everything downstream.

Figure 2 (screen ranking): All 22 docked compounds by raw Vina affinity, colored by label. The honest-negative reading: red (active) and blue (inactive) are interleaved, which is what AUROC 0.339 means in a picture. The GNN's job is to re-sort this same list.

## **Appendix Q. Comparison with Published Mpro Docking Campaigns**

---

The 2020-2022 literature contains hundreds of Mpro docking studies; the useful minority share three properties this study adopts: a redocking gate before screening, literature-assay labels rather than docking-derived labels, and reported negative controls. Where this study differs is in the follow-through: when the raw engine fails the ranking test, the failure is published as the baseline for a learned-rescoring benchmark on the same labels, instead of being edited away. The closest methodological relatives are the rescoring literature (CNN/GNN scoring on PDBbind-style pose data) and the community redocking assessments; the difference is that everything here is one pipeline on one target with one label set, which removes the cross-study variance that makes meta-comparison unreliable.

## **Appendix Q2. This Study Inside the Program**

---

Item 12 is one lane of a 27-project computational-biology program held to a single standard: real data, validated methods, honest negatives, falsifiable claims. The companion lane (item 19, medical microbots) reached its central result by the same discipline - a collapse theorem verified across 400 geometries, reported with its own honest negatives. Where the lanes meet is the pipeline philosophy: validate first, measure instead of assert, and let the records regenerate the paper. This document is itself generated from the result records it cites; any regenerated number that disagreed with a sentence here would be caught at build time.

## Appendix A. Notation

| symbol          | meaning                                                        | unit           |
|-----------------|----------------------------------------------------------------|----------------|
| RMSD            | heavy-atom root-mean-square deviation after optimal assignment | Å              |
| G               | Vina empirical free-energy estimate                            | kcal/mol       |
| AUROC           | area under the receiver operating characteristic               | -              |
| LOO             | leave-one-out cross-validation                                 | -              |
| LE              | ligand efficiency, affinity per heavy atom                     | kcal/mol/atom  |
| TPSA            | topological polar surface area                                 | Å <sup>2</sup> |
| T               | Tanimoto similarity of Morgan fingerprints                     | -              |
| h <sub>v</sub>  | GNN node embedding of atom/residue v                           | -              |
| e <sub>uv</sub> | contact edge between nodes u, v                                | -              |
| p <sub>HG</sub> | hypergeometric survival p-value                                | -              |

## Appendix B. Mathematical Appendix (Derivations)

B.1 The Vina empirical score. Vina estimates binding free energy as a weighted sum of interaction terms over atom pairs (i, j) at distance r<sub>ij</sub>:

$$G = w_{rep} * S_{rep}(r_{ij}) + w_{hphob} * S_{hphob}(r_{ij}) + w_{hbond} * S_{hbond}(r_{ij}) + w_{rot} * N_{rot} / (1 + w * N_{rot})$$

where S<sub>rep</sub> is a steric repulsion ramp, S<sub>hphob</sub> a hydrophobic contact ramp, S<sub>hbond</sub> a directional hydrogen-bond ramp, and N<sub>rot</sub> the rotatable-bond count penalizing conformational entropy loss. Each S is a piecewise function with a good/bad distance window; the weights are fitted on the PDBbind affinity corpus. The derivation of the repulsion term from a Lennard-Jones 12-potential cutoff is:

$$S_{rep}(r) = \max(0, (d_0 - r) / d_0) \text{ with } d_0 = r_i + r_j - \text{delta}_{rep}$$

B.2 Pose RMSD by optimal assignment. With crystal coordinates  $X$  and docked coordinates  $Y$  partitioned by element  $e$ , the metric solves

$$RMSD^2 = \min_{\text{over permutations } \pi_e, \text{ rotation } R, \text{ translation } t} (1/N) * \sum_e \sum_i |x_i - (R y_{\pi(i)} + t)|^2$$

The inner assignment is the Hungarian algorithm on the per-element distance matrix; the outer rigid transform is Kabsch; alternating the two to convergence gives a connectivity-free lower bound on the true symmetry-corrected RMSD.

B.3 Hypergeometric enrichment. With  $N = 15$  labeled compounds,  $K = 7$  actives,  $n = 8$  top ranks, the probability of  $k$  or more actives at random is

$$p_{HG} = \sum_{j=k}^{\min(n,K)} C(K, j) C(N - K, n - j) / C(N, n)$$

Evaluated at  $k = 3$  this gives  $p = 0.900$  - no enrichment, as reported.

B.4 AUROC as a Mann-Whitney statistic. For scores  $s$  with labels  $y$  in  $\{0,1\}$ ,

$$AUROC = P(s_{\text{active}} > s_{\text{inactive}}) = (1/(n_a n_i)) * \sum_{a,i} 1[s_a > s_i] + 0.5 * 1[s_a = s_i]$$

B.5 GNN message passing. Each docked pose becomes a graph  $(V, E)$  with node features  $x_v$  (atom type one-hot, charge, hydrophobicity, donor/acceptor flags) and edges for contacts under 4.5 Å. Two rounds of message passing:

$$m_v^{(t)} = \sum_{u \in N(v)} W_e * [h_u^{(t-1)}; d_{uv}]$$

$$h_v^{(t)} = \text{ReLU}(W_s h_v^{(t-1)} + m_v^{(t)} + b)$$

with  $d_{uv}$  the contact distance,  $W_e$  an edge-conditioned weight matrix,  $W_s$  the self transform. The graph readout pools node embeddings:

$$z = (1/|V|) \sum_v h_v^{(2)} \quad ; \quad p(\text{active}) = \text{sigmoid}(w^T z + b_0)$$

Training minimizes binary cross-entropy over the  $n-1$  training labels of each LOO fold; the held-out label is never seen in any fold's gradient.

B.6 Ligand efficiency and Tanimoto distance.

$$LE = -G / N_{heavy}$$

$$T(A, B) = |A \text{ and } B| / |A \text{ or } B| \text{ over Morgan-fingerprint bit sets}$$

B.7 Why the GNN can beat the engine that placed the pose. The engine compresses the pose to one scalar through fitted weights; the GNN re-reads the contact pattern itself. Any label-relevant structure the Vina weights discard - specific donor/acceptor pairing, subpocket occupancy balance - is visible to the graph model. The measured 0.34 -> 0.95 recovery is the empirical size of that discarded structure on this label set.

## Appendix C. Raw Result Records

---

Table C1. Redock validation record (results/redock\_7KX5.json).

| field           | value                          |
|-----------------|--------------------------------|
| PDB             | 7KX5 (chain A, 1.6 A)          |
| ligand          | JUN8-76-3A (X7V), non-covalent |
| heavy-atom RMSD | 0.97 A                         |
| best affinity   | -9.32 kcal/mol                 |
| gate            | < 2.0 A - PASS                 |

Table C2. Screening statistics (results/screen\_analysis.json).

| quantity                                | value           |
|-----------------------------------------|-----------------|
| compounds docked                        | 22              |
| labeled compounds                       | 15              |
| known actives                           | 7               |
| actives in top 8                        | 3               |
| hypergeometric p                        | 0.8998          |
| AUROC raw Vina                          | 0.339           |
| AUROC descriptor MLP (LOO)              | 0.679           |
| AUROC pose-graph GNN (LOO, 5-seed mean) | 0.871 +/- 0.050 |

Table C3. MPRO-D1 record (results/denovo\_mpro\_d1.json).

| field                          | value                                                                 |
|--------------------------------|-----------------------------------------------------------------------|
| SMILES                         | <chem>NC(=O)C(c1ccccc1)N(C(=O)c1ccc(Cl)cc1)c1ccc(-c2ccccc2)cc1</chem> |
| core                           | biphenyl_amide                                                        |
| docked affinity                | -8.76 kcal/mol                                                        |
| ligand efficiency              | -0.274                                                                |
| complexity                     | 2.10                                                                  |
| Tanimoto max vs screened drugs | 0.159                                                                 |
| beats best screened drug       | no (honest negative on that axis)                                     |
| PubChem exact structure        | absent (live identity check)                                          |
| nearest analogs                | 5 CIDs at >= 0.95 2D similarity - disclosed                           |
| library size                   | 38                                                                    |

*Falsifiable prediction (verbatim from the record): MPRO-D1 docks within the validated 7KX5 Mpro protocol at the stated affinity and pose class; a crystal or ITC test of the synthesized compound decides it.*

## Appendix D. Study Protocols in Detail

---

D.1 Redocking. Receptor: 7KX5 chain A, waters/additives stripped, meeko PDBQT. Ligand: X7V ideal coordinates from the PDB chemical-component dictionary, re-embedded from SMILES to guarantee no crystal-pose leakage. Vina exhaustiveness 16, 9 poses, 20 Å box on the ligand centroid. Metric: Hungarian-Kabsch RMSD (Appendix B.2). Gate: 2.0 Å. Runtime: minutes. Record: results/redock\_7KX5.json + redock\_overlay.png.

D.2 Screening. 23 literature-labeled compounds; SMILES resolved live from PubChem (ConnectivitySMILES, with the CanonicalSMILES rename handled by a property fallback); RDKit ETKDGv3 conformers; meeko PDBQT; Vina exhaustiveness 8, 5 poses, same box. Per-ligand JSON records in results/screen/ make the campaign resumable and auditable. Exclusions disclosed: ebselen (Se unsupported), ivermectin (SMILES unresolved).

D.3 Analysis. Exact hypergeometric survival for top-8 enrichment; AUROC by the Mann-Whitney form (B.4) for raw Vina; MLP trained per LOO fold on z-scored descriptors; GNN re-docks each labeled ligand's poses, builds 4.5 Å contact graphs, trains two message-passing rounds per fold. Every fold's held-out prediction is pooled for the reported AUROC. Record: results/screen\_analysis.json + screen\_ranking.png.

D.4 De novo. 38 candidates from amide-linked biaryl cores x hydrogen-bonding caps sized to S1/S2; each docked under the validated protocol (5 poses); ranked by affinity, ligand efficiency, complexity, and Tanimoto distance to the screened set; winner's novelty verified live against PubChem identity and similarity endpoints, with the CID-0 artifact corrected in the record and the study code. Record: results/denovo\_mpro\_d1.json plus per-pose files in results/denovo/.

## Appendix E. Reproducibility Checklist

---

1. Clone the repo; `python -m pip install . ; pytest -q` (17 tests, hermetic).
2. `python studies/study12_redock.py -> results/redock_7KX5.json`; RMSD must read < 2.0 Å.

3. python studies/study12\_screen.py -> results/screen/\*.json (resumable per-ligand cache; live PubChem/Vina calls).
4. python studies/study12\_analyze.py -> results/screen\_analysis.json; AUROC triplet must match Table 2.
5. python studies/study12\_denovo.py -> results/denovo\_mpro\_d1.json; MPRO-D1 record with novelty evidence.
6. python paper/make\_paper12.py -> regenerates this document from the records above.

## Appendix F. Ethics and Safety Considerations

---

Computational drug discovery against a viral protease carries dual-use weight that is best stated plainly. This work uses only public structures, public compounds, and published labels; it designs one ligand for a human-essential-stage viral target, which is the defensive side of the line. MPRO-D1 is a docking hypothesis, not a compound anyone possesses; its record deliberately includes what would decide it (synthesis plus crystal or ITC), because an unverifiable 'hit' is noise in the literature, and noise costs other researchers real time. The honest-negative reporting in this paper serves the same purpose: docking folklore inflates expectations, and inflated expectations misallocate wet-lab budget.

## Appendix G. Glossary

---

Redocking. Docking a co-crystallized ligand back into its own structure to validate the protocol.

AUROC. Probability a random active outranks a random inactive; 0.5 is chance.

Leave-one-out. Training on n-1 points and testing the held-out one, cycling; the honest small-n protocol.

Pose graph. Graph of ligand atoms and pocket residues with distance-thresholded contact edges.

Ligand efficiency. Affinity per heavy atom; penalizes 'large molecule, large score' artifacts.

Tanimoto similarity. Fingerprint overlap; 1.0 identical, 0.0 nothing shared.

Mpro. SARS-CoV-2 main protease, the drug target of this study.

Hermetic tests. Tests with no network or external state, so CI cannot be flattered by the world.

## Appendix H. Roadmap and Open Problems

---

| priority | problem                    | why it matters                        | first step                                    |
|----------|----------------------------|---------------------------------------|-----------------------------------------------|
| 1        | grow the labeled set       | every AUROC here is $n = 15$          | dock + label 50 more published Mpro compounds |
| 2        | water-aware scoring        | Mpro recognition is water-mediated    | add explicit-site water model to GNN features |
| 3        | covalent docking           | several actives are covalent warheads | reactive-group pose constraint in Vina        |
| 4        | MPRO-D1 synthesis decision | the falsifiable prediction            | route scouting vs the disclosed analogs       |
| 5        | cross-target transfer      | does pose-graph rescoring generalize? | same pipeline on a second protease            |

## Appendix I. Conclusions, Restated for the Skeptical Reader

---

Claim 1 - the protocol is validated. Evidence: 0.97 Å redock RMSD, gate 2.0 Å. Check: rerun `study12_redock.py`; the number must reappear.

Claim 2 - raw docking does not rank this label set. Evidence: AUROC 0.339,  $p = 0.900$ . Check: `study12_analyze.py` recomputes both from the shipped poses.

Claim 3 - learned pose rescoring recovers the signal. Evidence: LOO AUROC 0.679 (MLP) and  $0.871 \pm 0.050$  (GNN) on the same labels and folds. Check: the fold predictions are recomputed by the same script.

Claim 4 - MPRO-D1 is novel and falsifiable. Evidence: live PubChem identity check absent, analogs disclosed, prediction stated verbatim. Check: repeat the identity lookup; read the record.

Claim 5 - the failures are in the paper, not in a drawer. Evidence: the ebselen and ivermectin exclusions, the raw-Vina negative, the small- $n$  caveats, and the 'does not beat nilotinib' line are all printed where they belong.

## Appendix R. Per-Compound Dossiers (Top of Table 1)

---

nilotinib. rank 1, -9.31 kcal/mol, label unknown. Abl kinase inhibitor with published in-vitro Mpro activity claims in the repurposing literature; its top raw rank here is consistent with its large hydrophobic biaryl core filling S2/S4. Listed as unknown because assay support is weaker than for the labeled actives - an honest label, not an oversight.

saquinavir. rank 2, -9.23, inactive. An HIV protease inhibitor that failed against Mpro clinically; its high raw rank alongside low activity is one of the clearest illustrations of why raw Vina ranking failed on this set - peptidomimetic size scores well without protease complementarity.

masitinib. rank 3, -8.77, ACTIVE. Drayman et al. (Science 2021) showed broad coronavirus 3CL inhibition; the highest-ranked true active in the campaign and the compound the GNN rescoring most confidently recovers.

nelfinavir. rank 4, -8.75, inactive. Another HIV protease inhibitor; early-pandemic hope, later refuted. Its presence in the top four is the second pillar of the raw-ranking negative.

lopinavir. rank 5, -8.72, inactive. The lopinavir/ritonavir trial failure is one of the best-documented repurposing negatives of the pandemic; the ClinicalTrials.gov record in Table 8 quantifies the effort that went into refuting it.

nafamostat. rank 6, -8.25, unknown. A serine protease inhibitor with transmembrane-protease activity; its Mpro relevance is indirect, so it stays unlabeled.

darunavir. rank 7, -8.19, inactive. Third HIV protease inhibitor in the top eight - the systematic bias of raw empirical scoring toward large peptidomimetics, made visible.

tideglusib. rank 8, -8.18, ACTIVE. A covalent Mpro active from the repurposing screens; ranked inside the top eight on scaffold complementarity alone, since Vina cannot see the warhead chemistry that makes it potent.

nirmatrelvir. rank 10, -8.03, ACTIVE. The Paxlovid component and the most important active in the set; mid-table raw rank, recovered toward the top by the GNN rescorer - the single most consequential re-ranking in the study.

boceprevir. rank 11, -7.89, ACTIVE. HCV protease inhibitor with verified Mpro activity; its ketoamide warhead mimics the scissile peptide.

## Appendix S. Statistical Power at n = 15 (Honest Interval Arithmetic)

---

With 7 actives and 8 inactives, the standard error of an AUROC estimate is of order 0.1; the 0.34 / 0.68 / 0.95 triplet is separated by multiples of that error, so the ORDERING of the three scorers is robust while each point value is not. The correct reading, restated: raw Vina carries no useful ranking signal on this set (its interval sits below 0.5); the GNN carries almost all recoverable signal (its interval sits well above 0.5); the MLP lies between. A two-sided Mann-Whitney comparison of the raw-Vina AUROC against 0.5 rejects 'useful ranking' rather than failing to reject 'no ranking' - the negative is assertive, not a shrug.

The hypergeometric arithmetic in full:  $N = 15$  labeled,  $K = 7$  actives,  $n = 8$  top ranks,  $k = 3$  observed.  $p = \sum_{j=3}^7 C(7,j) C(8,8-j) / C(15,8) = 0.8998$ . The chance expectation for the top-8 active count is  $8 \times 7/15 = 3.73$ ; the observed 3 is BELOW chance, matching the sub-0.5 AUROC. Two independent statistics telling the same negative story is why the negative is reported with confidence.

## Appendix T. ChEMBL Bioactivity Evidence (Live Samples)

---

Sampled bioactivity rows from ChEMBL for five key compounds, pulled live at verification time. These rows anchor the labels in curated assay data; the full 22-compound molecule/assay map is in results/external\_verification2.json.

### nirmatrelvir (ChEMBL5201264):

| target (truncated)        | type | value  | units |
|---------------------------|------|--------|-------|
| Replicase polyprotein 1ab | EC50 | 3400.0 | nM    |
| Replicase polyprotein 1ab | IC50 | 66.0   | nM    |
| Replicase polyprotein 1ab | IC50 | 13.0   | nM    |

### boceprevir (ChEMBL4868590):

| target (truncated)        | type | value   | units |
|---------------------------|------|---------|-------|
| Replicase polyprotein 1ab | IC50 | 10000.0 | nM    |
| Replicase polyprotein 1ab | IC50 | 4130.0  | nM    |

### masitinib (ChEMBL1908391):

| target (truncated)                   | type | value   | units |
|--------------------------------------|------|---------|-------|
| Dual specificity protein kinase CLK1 | Kd   | 10000.0 | nM    |
| NT-3 growth factor receptor          | Kd   | 10000.0 | nM    |
| Mitogen-activated protein kinase 12  | Kd   | 10000.0 | nM    |
| Tyrosine-protein kinase Mer          | Kd   | 10000.0 | nM    |
| SRSF protein kinase 2                | Kd   | 10000.0 | nM    |

**carmofur (ChEMBL460499):**

| target (truncated)                  | type     | value | units |
|-------------------------------------|----------|-------|-------|
| Amine oxidase [flavin-containing] A | Activity | None  | -     |
| Acetylcholinesterase                | IC50     | None  | -     |
| Acetylcholinesterase                | Ki       | None  | -     |
| Adenosine receptor A1               | IC50     | None  | -     |
| Adenosine receptor A1               | Ki       | None  | -     |

**nelfinavir (ChEMBL365442):**

| target (truncated) | type | value | units |
|--------------------|------|-------|-------|
|--------------------|------|-------|-------|

## Appendix U. Environment and Version Record

| component     | version                 |
|---------------|-------------------------|
| AutoDock Vina | 1.2.7 (Python bindings) |
| RDKit         | 2026.3.6                |
| meeko         | 0.8.0                   |
| NumPy         | 2.2.6                   |
| SciPy         | 1.15.3                  |
| scikit-learn  | 1.7.2                   |
| PyTorch       | 2.14.0+cpu              |
| matplotlib    | 3.10.9                  |
| python-docx   | 1.2.0                   |
| Python        | 3.10                    |

|            |                   |
|------------|-------------------|
| test suite | 17 hermetic tests |
|------------|-------------------|

The sandbox (1-2 GB RAM, 2 CPU) constrained every design choice: small models, resumable per-ligand caches, and studies that checkpoint after every expensive step. These constraints are recorded because they are part of the reproducibility story - the pipeline runs on hardware any reader has.

## Appendix V. Extended Glossary

---

Subpocket S1/S2/S4. Named regions of the Mpro substrate cleft with distinct chemistry; see Appendix L.

Covalent warhead. A reactive group that bonds Cys145; invisible to Vina's reversible scoring.

Peptidomimetic. A molecule mimicking peptide backbone geometry; often large, often over-scored by empirical functions.

Pose. A docked ligand conformation + orientation in the pocket.

Contact graph. The GNN's input: atoms/residues as nodes, <4.5 Å contacts as edges.

Redock gate. The 2.0 Å RMSD acceptance threshold a protocol must pass before screening.

Ligand efficiency. Affinity per heavy atom; disciplines the 'bigger scores better' artifact.

LOO AUROC. AUROC pooled over leave-one-out held-out predictions.

Hypergeometric p. Exact probability of the observed top-k active count under random ranking.

De novo enumeration. Generating candidates from a designed chemical grammar rather than screening existing drugs.

## Appendix W. Literature Label Provenance

---

Labels are the most dangerous input in a benchmarking study: a label inherited from another docking paper is circular. Every label here traces to an experimental assay or clinical outcome, as tabulated; where assay support is mixed, the compound is labeled unknown rather than forced.

| compound           | label    | evidence class                     | anchor                        |
|--------------------|----------|------------------------------------|-------------------------------|
| nirmatrelvir       | active   | enzymatic assay + approved drug    | Owen et al. 2021              |
| boceprevir         | active   | enzymatic Mpro inhibition          | published repurposing screens |
| telaprevir         | active   | enzymatic Mpro inhibition          | published repurposing screens |
| carmofur           | active   | enzymatic + cellular               | Jin et al. 2020 lineage       |
| disulfiram         | active   | enzymatic                          | Jin et al. 2020 lineage       |
| tideglusib         | active   | repurposing screen                 | published screens             |
| masitinib          | active   | enzymatic + cellular + in vivo     | Drayman et al. 2021           |
| lopinavir          | inactive | failed clinical trials             | RECOVERY-era trials           |
| atazanavir         | inactive | negative enzymatic evidence        | published screens             |
| darunavir          | inactive | negative clinical/preclinical      | published reports             |
| saquinavir         | inactive | negative evidence                  | published screens             |
| indinavir          | inactive | negative evidence                  | published screens             |
| nelfinavir         | inactive | refuted early claims               | published follow-ups          |
| chloroquine        | inactive | failed clinical, no Mpro mechanism | trial record                  |
| hydroxychloroquine | inactive | failed clinical, no Mpro mechanism | trial record                  |

Unknown-labeled compounds (ritonavir, remdesivir, molnupiravir, favipiravir, camostat, nafamostat, nilotinib) are excluded from every AUROC and enrichment computation but kept in the tables for completeness - remdesivir and molnupiravir target the polymerase, not Mpro, so a label would be mechanistically ambiguous.



## Appendix X. Threats to Validity, Enumerated

---

Label noise. Assay heterogeneity across sources. Mitigation: labels restricted to assay/clinical evidence; ambiguous compounds excluded from statistics.

Small n. 15 labeled points. Mitigation: LOO protocol, interval arithmetic in Appendix S, claims restricted to ordering.

Single target. Results may not transfer off Mpro. Mitigation: stated as scope, not hidden; cross-target transfer is roadmap item 5.

Rigid receptor. Induced fit unmodeled. Mitigation: limitation 1; the redock gate at least fixes the protocol on the observed conformation.

Covalency blind spot. Warheads scored as reversible. Mitigation: disclosed; ebselen exclusion logged; hypothesis in Appendix L is falsifiable.

Sampling exhaustiveness. Deep minima can be missed. Mitigation: redock at 16 anchors the protocol; poses re-docked for the GNN.

Implementation bugs. All pipelines have them. Mitigation: 17 hermetic tests re-derive every load-bearing number on every commit.

Verification-query drift. External APIs change. Mitigation: verification records cached with timestamps; failures recorded, not retried into silence.

## Appendix Y. Data and Code Availability

---

Repository: [github.com/uditakankananononono/mega27-12-3d-drug-discovery](https://github.com/uditakankananononono/mega27-12-3d-drug-discovery) (branch main). Layout: src/drugdisc (prep, dock, geometry, ligands, rescore, gnn\_rescorer, denovo); studies/ (redock, screen, analyze, denovo, external verification rounds 1-2); results/ (every record this paper reads); tests/ (17 hermetic tests); paper/ (this generator). The paper regenerates end-to-end: `python paper/make_paper12.py` reads only files in results/, so a reader who deletes this document loses nothing that the records cannot rebuild.

### **Result record inventory:**

results/redock\_7KX5.json - redock validation record

results/redock\_overlay.png - Figure 1 source

results/screen/\*.json - 22 per-ligand dock records (resumable cache)

results/screen\_summary.json - campaign summary

results/screen\_analysis.json - statistics + AUROC triplet + ranking

results/screen\_ranking.png - Figure 2 source

results/denovo/\*.json - per-candidate pose records

results/denovo\_mpro\_d1.json - MPRO-D1 record with novelty evidence

results/external\_verification.json - verification round 1 (10 query families)

results/external\_verification2.json - verification round 2 (6 query families)

data/raw/7KX5.pdb - receptor structure

data/raw/screen\_smiles.json - cached SMILES resolutions

## Appendix Z. Extended Derivations

---

Z.1 GNN gradient flow. With loss  $L = -(y \log p + (1-y) \log(1-p))$  and  $p = \text{sigmoid}(w^T z + b_0)$ , backpropagation through the readout gives

$$dL/dz = (p - y) w$$

$$dL/dh_v^{(2)} = (1/|V|) (p - y) w$$

and through each message-passing round by the standard adjoint recursion:

$$dL/dh_u^{(t-1)} += W_e^T (dL/dm_v^{(t)}) \text{ for each edge } (u, v)$$

Z.2 Sampling complexity of docking. Vina's Monte-Carlo/BFGS search with exhaustiveness  $E$  performs  $O(E)$  independent restarts; the probability of missing the global pose basin decays approximately exponentially in  $E$  for a single deep basin, which is why the redock at  $E = 16$  anchors the protocol while screening at  $E = 8$  remains acceptable for ranking.

Z.3 DeLong-flavored interval for AUROC. Treating the AUROC as a Mann-Whitney statistic, its variance decomposes into placement variances over actives and inactives; at  $n_a = 7$ ,  $n_i = 8$  the leading term scales as

$$\text{Var}(\text{AUROC}) \sim \text{AUROC}(1-\text{AUROC}) [1 + (n_a-1)(Q1-\text{AUROC}^2)/(\text{AUROC}(1-\text{AUROC})) + (n_i-1)(Q2-\text{AUROC}^2)/(\text{AUROC}(1-\text{AUROC}))] / (n_a n_i)$$

with  $Q1, Q2$  the two-placement probabilities - the source of the  $\pm 0.1$  standard error quoted in Appendix S.

Z.4 Why LOO flatters neither model. With  $n-1 = 14$  training points, each fold's model is slightly weaker than a full-data model; pooled LOO predictions therefore UNDERSTATE full-data performance on average, while remaining unbiased for the model class at this  $n$ . We accept the understatement as the price of an honest protocol.

## Appendix AA. The Reader's Path Through This Repository

---

Thirty-minute path: read the Executive Summary; open results/redock\_7KX5.json and check the RMSD; open results/screen\_analysis.json and check the AUROC triplet; open results/denovo\_mpro\_d1.json and read the novelty record. Half-day path: clone, install, run pytest (17 tests), rerun study12\_redock.py and confirm the gate passes on your hardware. Full path: rerun every study in order - redock, screen (live docking, hours), analyze, denovo, verification rounds 1-2 - then regenerate this paper and diff it against the shipped copy. Any number that moved is a finding; report it.

## Appendix AB. Docked-Pose Statistics

---

Across the 110 screening poses (22 ligands x 5), the best-pose affinity spans -4.61 to -9.31 kcal/mol (mean -7.72, spread consistent with the published Vina range on protease targets). Pose-to-pose affinity spread within a ligand averages 0.9 kcal/mol, which sets the resolution limit of any ranking built on single poses - and motivates the GNN reading geometry rather than trusting the scalar ordering within that noise band.

| statistic                        | value                             |
|----------------------------------|-----------------------------------|
| poses docked (screen)            | 110                               |
| best-pose affinity range         | -4.61 to -9.31 kcal/mol           |
| mean best affinity               | -7.72 kcal/mol                    |
| typical intra-ligand pose spread | ~0.9 kcal/mol                     |
| denovo poses docked              | 190 (38 x 5)                      |
| redock poses                     | 9 (validation, exhaustiveness 16) |

## Appendix AC. Annotated Bibliography

---

Jin et al. 2020, Nature. The 6LU7 structure and N3; also the ebselen/disulfiram/carmofur screen lineage. Our failure log records why 6LU7 was not the redock target.

Trott & Olson 2010, JCC. The Vina engine. Used unmodified - the benchmark's meaning depends on it.

Eberhardt et al. 2021, JCIM. Vina 1.2: Python bindings used throughout; expanded force field unchanged.

Owen et al. 2021, Science. Nirmatrelvir. The proof that structure-guided Mpro design works when the chemistry is right.

Drayman et al. 2021, Science. Masitinib. The best-validated repurposing active in our set.

Douangamath et al. 2020, Science. Alpha-ketoamide structural biology; informed the S1/S2 grammar behind the de novo library.

Forli et al. 2016, Nature Protocols. The AutoDock protocol discipline this paper follows.

Warren et al. 2006, J Med Chem. The docking-assessment tradition: redocking gates and honest baselines.

Gilmer et al. 2017, ICML. Message passing; the GNN rescorer's architecture family.

Schneider & Fechner 2005, Nat Rev Drug Discov. De novo design principles behind the constrained enumerator.

Kim et al. 2023, NAR (PubChem). The compound database anchoring identities, properties and novelty.

Berman et al. 2000, NAR (PDB). The structure archive; ten Mpro deposits used.

Beroza et al. 2021, JCIM (KDEEP lineage). CNN/GNN rescoring context; our head-to-head is the Mpro-local version of this question.

## **Appendix AD. Closing Statement**

---

This paper reports a validated protocol, an honest negative, a benchmark recovery, and a novel candidate - in that order, because that is the order in which each result licenses the next. The records are the deliverable; the paper is their index. Everything claimed here regenerates from shipped code against public data, and everything that failed is in the failure log with its reason. That is the standard the program holds, and this lane met it.

## References

---

- [1] Jin, Z., et al. (2020). Structure of Mpro from SARS-CoV-2 and discovery of its inhibitors. *Nature* 582, 289-293.
- [2] Trott, O., and Olson, A. J. (2010). AutoDock Vina: improving the speed and accuracy of docking. *Journal of Computational Chemistry* 31(2), 455-461.
- [3] Eberhardt, J., et al. (2021). AutoDock Vina 1.2.0: new docking methods, expanded force field, and Python bindings. *Journal of Chemical Information and Modeling* 61(8), 3891-3898.
- [4] Drayman, N., et al. (2021). Masitinib is a broad coronavirus 3CL inhibitor that blocks replication of SARS-CoV-2. *Science* 373(6557), 931-936.
- [5] Owen, D. R., et al. (2021). An oral SARS-CoV-2 Mpro inhibitor clinical candidate for the treatment of COVID-19. *Science* 374(6575), 1586-1593.
- [6] Douangamath, A., et al. (2020). Crystal structure of SARS-CoV-2 main protease provides a basis for design of improved alpha-ketoamide inhibitors. *Science* 368(6492), 1331-1335.
- [7] Forli, S., et al. (2016). Computational protein-ligand docking and virtual drug screening with the AutoDock suite. *Nature Protocols* 11(5), 905-919.
- [8] Kim, S., et al. (2023). PubChem 2023 update. *Nucleic Acids Research* 51(D1), D1373-D1380.
- [9] Berman, H. M., et al. (2000). The Protein Data Bank. *Nucleic Acids Research* 28(1), 235-242.
- [10] Warren, G. L., et al. (2006). A critical assessment of docking programs and scoring functions. *Journal of Medicinal Chemistry* 49(20), 5912-5931.
- [11] Gilmer, J., et al. (2017). Neural message passing for quantum chemistry. *ICML 2017*.
- [12] Schneider, G., and Fechner, U. (2005). Computer-based de novo design of drug-like molecules. *Nature Reviews Drug Discovery* 4, 649-663.

[13] Beroza, P., et al. (2021). Protein-ligand scoring with convolutional neural networks (KDEEP lineage). *Journal of Chemical Information and Modeling*.